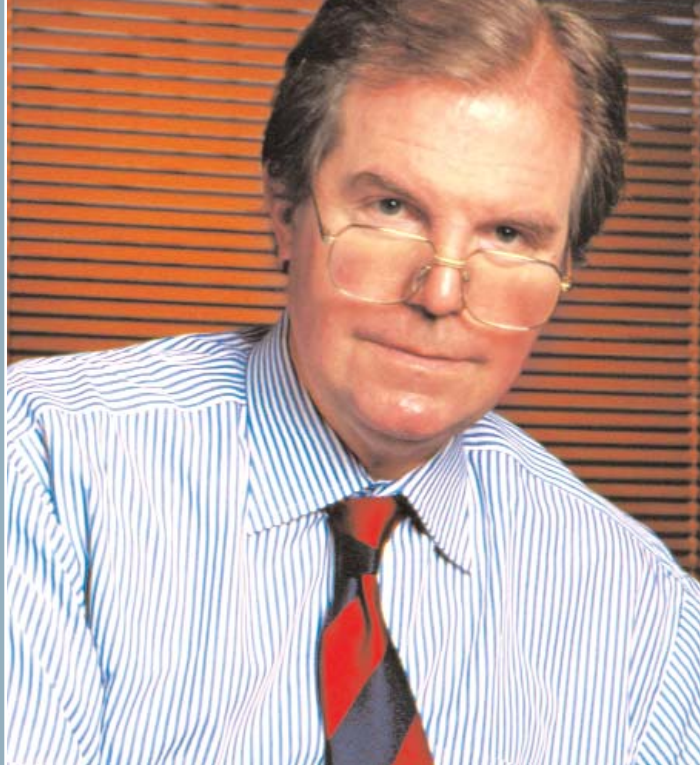




“Computing is not about computers any more. It is about living.”

Nicholas Negroponte is founder and director of the Media Laboratory at the Massachusetts Institute of Technology, Cambridge, USA

Of Bits, Atoms, And Softwear

One of the first definitions of convergence came from Nicholas Negroponte: “Bits co-mingle effortlessly. They start to get mixed up and can be used and re-used separately or together. The mixing of audio, video, and data is called multimedia. It sounds complicated, but it’s nothing more than co-mingled bits.”

Negroponte pointed out in his 1995 bestseller *Being Digital* that our world is fast exchanging the trade in atoms for the trade in bits. In other words, he has been a big proponent of the ‘bit switch’—the idea that we’re moving from producing and distributing atoms in the form of products to producing digital things in the form of bits that get pushed around the world.

We present two columns Negroponte wrote for *Wired* magazine—‘Wearable Computing’ and ‘Bits And Atoms’. These columns read as well today as they did when they were first published.

Wearable Computing

The digital road warrior’s kit—laptop, cell phone, PDA, and pager—is just capable enough to bother you everywhere without necessarily helping you anywhere. It’s absurd that each device is still on such poor speaking terms with the others. We walk around like pack horses saddled with information appliances. We should be in the saddle, not under it.

The Evolution Of Softwear

More than 20 years ago, The Architecture Machine Group at MIT built a mediaroom based on the idea that one should be inside a computer rather than in front of it. While that vision foreshadowed today’s immersive environments, it did not go far enough and shrink the room to the size of a person.

In the future, the PC will be blown to bits, many of which, naturally, should be kept near you rather than in your home or at your office. But so far, software has not been particularly soft. Though bits are as insubstantial as the ether, they tend to be packaged in hard boxes. For hardware and software to comfortably follow you around, they must merge into softwear. Developing wearable computing requires as much attention to the medium as the message.

In fact, the medium becomes the message.

What single manufactured material are you exposed to the most? The answer is fabric. We wear it, stand on it, sit on it, and sleep in it. Marvellous technology goes into looms, but all we ask fabric to do is protect us from the elements, look pretty, and not wrinkle or shrink. Can’t it do more?

Advances in conducting polymers and reversible optical media are pointing towards

fabrics that can literally become displays. Amorphous semiconductors can be used to make solar cells to power fabric. Polymer semiconductors are candidates for wearable logic. The result would be the ultimate flexible computer architecture. Fashion accessories will take on new roles, becoming some of the most important Internet access points, conveniently surrounding you in a Person Wide Web. How better to receive audio communications than through a earring, or to send spoken messages than through your lapel?

Footwear is particularly attractive for computing. Your shoes have plenty of unclaimed space, receive an enormous amount of power (from walking) that is currently untapped, and are ideally placed to communicate with your body and the ground. When you come home, your shoes can talk to the carpet in preparation for delivery of the day’s personalised news to your glasses.